

Hospital Heroes

Junior Doctor · Children's Hospital

15 SHIFTS · 58 STOPS · GRADE 2 · HOSPITAL

THE OPENING CARD

You are a new doctor at the children's hospital. Today is your first day on the ward. Children come in hurt, or sick, or scared. Some of them are too little to say what is wrong. Nurse Alex Lee will show you round this morning. Then she will start asking you what you saw. Your job is to look closely, ask good questions, and work out who needs help first. Three children are waiting by the door right now.

Morning Rush

BEFORE THE SHIFT — GREET

Say hello to the team before your first shift

Lots of people work on this floor. Doctors, nurses, and the people who bring the food. You will need to ask them things all week. Go and meet as many of them as you can. It is easier to ask somebody a question when you know their name.

PLAN CARD 76W

It is your first shift. Three children come in at once. Ava is quiet and breathing hard. Ben is crying loudly about his ankle. Mia has not said anything at all. Nurse Alex Lee, who is showing you the ward today, says to watch all three before you choose. Today you decide who to see first, and you learn the three kinds of clue. The loudest patient is not always the one in the most trouble.

BRIEFING 10W

How doctors use clues and decide who needs help first

1.1 Who Needs You First?

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A ROOM · TRIAGE

SCENE 65W

Three patients arrive at almost the same time. Ava sits very still and breathes hard. Ben can walk, but his ankle hurts. Maya has an itchy patch on her arm. One exam room is free. Hospitals do not always help people in the order they arrive. Some problems stay the same for an hour. Others get worse fast. Picking who goes first is called triage.

GUIDE 42W

All four people need help. That is not the question. Ask one thing of each. Which of them gets worse while you help somebody else? A sore ankle stays about the same. So does an itchy patch. Breathing is not like that.

ASK 10W

Which patient should move to the front of the line?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

your body needs air all the time

VERDICT 52W

Breathing brings oxygen into the body. Every part needs it, all the time. So when breathing is hard, the whole body is in trouble fast. A sore ankle hurts and stays about the same. An itchy patch stays about the same too. You go to the one that can get worse quickest.

TAKEAWAY 13W

Doctors use urgency, not first-come-first-served order, when several patients need help at once.

1.2 Observation or Measurement?

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A PERSON · CHOICE

SCENE 59W

Luis says his belly hurts. You can see that he is awake and talking. A nurse checks his temperature with a thermometer. She counts his pulse too. All of it goes on his chart. But the clues are not all the same kind. Some you notice. Some the patient tells you. Some come from a tool, as a number.

GUIDE 41W

All four clues are real and useful. They come from different places. One you see. One the patient tells you. One comes from a tool, as a number. Ask of each where it came from. That is what decides this one.

ASK 5W

Which item is a measurement?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a tool can give you a number

VERDICT 56W

Clues come in three kinds. Some you see, like a worried face. Some the patient tells you, like where it hurts. Some come from a tool, as a number. A thermometer gives a number. That makes it a measurement. All three kinds are useful. Knowing which kind you have tells you how sure you can be.

TAKEAWAY 13W

Doctors combine observations, patient stories, and measurements. They are different kinds of clues.

1.3 Your First Exam Room

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A PERSON · SEQUENCE

SCENE 29W

Ben is here for his ankle check. The team has four things to do. Clean hands. Ben's story. A look at the ankle. A note for the next person.

GUIDE 38W

All four steps happen. So ask what each one needs first. What is on your hands goes on to Ben. His story tells you where to look. And the note is for the nurse who comes after you.

ASK 7W

Put these steps in the best order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

germs can travel on hands

VERDICT 44W

Each step makes the next one work. Hands come first. What is on them goes on to Ben. His story comes next. It tells you where to look. Then you look. Then you write it down, so the next person knows what you saw.

TAKEAWAY 14W

Each step in a hospital routine is there to make the next one work.

END OF THE SHIFT 2W

Shift complete

Lunch Shift

BEFORE THE SHIFT – TRIAL

Walk the whole floor one time

The rooms all look the same from the hall. Walk past every one now, while it is quiet. Later somebody will need you in a hurry. You want to know which way to go without stopping to read the signs.

PLAN CARD 76W

Second shift, just after lunch. Owen ate a sandwich an hour ago and his tummy hurts. He points to the middle of his tummy, and then a bit lower down. He had the same sandwich yesterday and was fine. Nurse Alex Lee wants to know what he ate and when. Today you follow food through the body and work out which part does which job. Each part is built for one job, and the order matters.

BRIEFING 7W

Digestion and how food becomes usable fuel

2.1 Follow the Sandwich

FOOD & WATER · NUTRITION & KIDNEY CARE · A ROOM · SEQUENCE

SCENE 31W

Owen has a paper sandwich card. He wants to know where it goes after he swallows. The dietitian hands you four organ cards and asks you to put them in order.

GUIDE 40W

These four are one journey, and each part has one job. Ask what has to happen first. Nothing can soak up tiny bits before something has made them tiny. Food is far too big for the body to use whole.

ASK 6W

Put the food journey in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

food gets chewed before you swallow it

VERDICT 57W

Food is far too big for the body to use. It gets broken down in stages. The mouth chews it and wets it. It goes down the food tube. The stomach mixes it and breaks it down more. Then the small intestine takes the tiny bits into the blood. Each part does one piece of the job.

TAKEAWAY 9W

Different organs have different jobs in one connected system.

2.2 Why Food Is Fuel

FOOD & WATER · NUTRITION & KIDNEY CARE · A PERSON · CHOICE

SCENE 26W

Nina ran all morning and now she feels shaky. She has not eaten since breakfast. Her nurse asks you what her body has run out of.

GUIDE 39W

Four reasons Nina might feel shaky. Ask of each whether the story fits. She ran all morning. She has not eaten. Two of these answers are about water or about nerves. Food is fuel, and running spends it fast.

ASK 13W

Nina ran all morning and has not eaten. Why does she feel shaky?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

food gives your body energy

VERDICT 63W

Food is fuel. Your body breaks food down and uses the energy to move, to keep warm, and to think. Running uses a lot of energy fast. Nina ate at breakfast and then used it all up. When the fuel runs low, people feel shaky, tired, or grumpy. A snack puts the fuel back. This is why hospitals ask when someone last ate.

TAKEAWAY 15W

The body spends what it takes in, and spends it faster when it works harder.

2.3 Three Jobs, Three Organs

FOOD & WATER · NUTRITION & KIDNEY CARE · A PERSON · CHOICE

SCENE 30W

Three cards each need one picture. Owen needs the part that starts breaking food up. Sam needs the part that mixes it. Nia needs the part that soaks it up.

GUIDE 38W

Four sets, and only one has every job in the right place. Ask what each part does. Chewing makes pieces smaller. Mixing breaks them down more. Soaking up comes last. Check all three before you pick a set.

ASK 5W

Which matching set is correct?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

different body parts do different jobs

VERDICT 47W

A body system works like a good team. Nobody does everything. The mouth chews, so the pieces get smaller. The stomach mixes and breaks them down more. The small intestine soaks up the tiny bits. Each part is built for its own job, and the order matters.

TAKEAWAY 14W

A body system works because specialized parts each do a piece of the job.

END OF THE SHIFT 2W

Shift complete

X-Ray Day

BEFORE THE SHIFT — FOLLOW

Walk with Nurse Alex on the round

Nurse Alex checks on every child, in the same order, every morning. Walk along behind and watch. You will see where she stops and what she looks at. That is how the round is learned.

PLAN CARD 75W

Third shift, and Ben is back — the boy with the sore ankle from your first day. He remembers you and is trying not to cry again. This time it is his wrist, and it is swollen. He fell onto his hand from the monkey bars an hour ago. Dr. Elena Garcia is waiting for you. Today you pick the right tool, learn what bones do, and keep Ben calm while the team gets ready.

BRIEFING 6W

Bones, joints, protection, and medical imaging

3.1 Why Take an X-Ray?

BONES, MUSCLES & SKIN · IMAGING & REHAB · A ROOM · PROTOCOL

SCENE 34W

Eli fell off his bike. His wrist hurts and looks swollen. The doctor needs to know if a bone is broken. The bone is under the skin, and nobody can see it from outside.

GUIDE 41W

Three clues, and three tools. Match each clue to the tool that can find it. Every tool here is useful, and each one answers a different question. The one that finds a broken bone cannot tell you how hot somebody is.

ASK 12W

Match each clue the doctor needs to the tool that finds it.

BEHIND THE BUTTON 96W

What an X-ray does. It makes a picture of the hard parts inside, because bone stops the rays and soft parts do not. That is why it can show a crack nobody can see from outside.

What the other tools do. A thermometer measures how hot somebody is. A stethoscope lets you listen to sounds inside, like the heart and the lungs.

Why matching is the skill. A hospital has many tools and none of them answers every question. Choosing the one that fits the clue you need is most of what a good team does.

TAKES AS READ

some things are hidden under the skin

VERDICT 94W

Every tool here is useful. Each one answers a different question. Bone stops X-rays. Soft parts do not. So an X-ray shows the hard parts inside. That is the only way to see if the bone in Eli's wrist is cracked. A thermometer tells you how hot he is. That says nothing about a broken bone. A stethoscope lets you listen inside him. That is how you check both lungs. Picking the right tool is most of the job. The wrong tool still gives you an answer. It just answers a question nobody asked.

TAKEAWAY 11W

Doctors choose tools based on the kind of clue they need.

3.2 Skeleton Jobs

BONES, MUSCLES & SKIN · IMAGING & REHAB · A PERSON · CHOICE

SCENE 63W

The bone doctor has three cards to hand out. Rosa needs the part that protects the heart and lungs. Mr. Bell needs the part that holds up the back. Tess needs the place where bones meet and move. Bones do more than hold you up. They are not all the same shape. That is because they do not all do the same job.

GUIDE 38W

Four sets, and bones do more than one job. Ask what each bone is shaped for. A cage guards what is inside it. A column holds you up. A joint is where two bones meet, so it bends.

ASK 4W

Which set is correct?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

bones are hard and do not bend

VERDICT 55W

Bones do more than one job. The rib cage is a cage, so it guards the heart and lungs inside it. The spine holds you up and guards the nerve cord running down your back. A joint is a place where two bones meet, and that is where you bend. Same material, three different jobs.

TAKEAWAY 9W

The skeleton supports, protects, and helps the body move.

3.3 Possible Broken Bone

BONES, MUSCLES & SKIN · IMAGING & REHAB · A PERSON · SEQUENCE

SCENE 27W

Eli's wrist may be hurt. He looks worried. In this game you do not fix it. You keep him safe and get the team what they need.

GUIDE 41W

All four steps happen. Ask which one stops things getting worse. Everything else can happen while the wrist is held still. Help takes time to come, so telling somebody is early. The note comes last, and it holds all of it.

ASK 7W

Put the safe hospital response in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

moving something that is hurt can make it worse

VERDICT 46W

Keeping the wrist still comes first. It is the only step that stops more harm. All the rest can happen while you hold it still. Then you tell a grown-up, because help takes time to come. The note goes last. It writes down all of it.

TAKEAWAY 18W

The first action is the one that stops things getting worse; the rest can happen while it holds.

3.4 Who Needs You First? — Review 2

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A ROOM · CALLBACK · TRIAGE

SCENE 51W

Three patients arrive at almost the same time. Ava is sitting very still and breathing hard. Ben can walk, but his ankle hurts. Maya is smiling and showing the nurse an itchy patch on her arm. The team asks you to decide who should be brought to an exam room first.

GUIDE 42W

All four people need help. That is not the question. Ask one thing of each. Which of them gets worse while you help somebody else? A sore ankle stays about the same. So does an itchy patch. Breathing is not like that.

ASK 10W

Which patient should move to the front of the line?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

your body needs air all the time

VERDICT 52W

Breathing brings oxygen into the body. Every part needs it, all the time. So when breathing is hard, the whole body is in trouble fast. A sore ankle hurts and stays about the same. An itchy patch stays about the same too. You go to the one that can get worse quickest.

TAKEAWAY 13W

Doctors use urgency, not first-come-first-served order, when several patients need help at once.

END OF THE SHIFT 2W

Shift complete

Brain Signal Day

PLAN CARD 74W

Fourth shift. Arjun bumped his head at break. His knee is scraped and you can see it. He is also muddled about what day it is, and you can only see that if you talk to him. His friend says he asked the same question twice on the way in. Dr. Sam Kim asks which one worries you. Today you follow signals into the brain and back out, and decide who needs help first.

BRIEFING 7W

Brain, spinal cord, nerves, and fast responses

4.1 The Body's Slow Messages

BRAIN & SENSES · NEUROLOGY & SENSES · A ROOM · CHOICE

SCENE 30W

Theo is 11 and has grown 4 centimetres since spring. His chart shows he grew most while he was asleep. He wants to know who told his bones to grow.

GUIDE 40W

Four things that might have told his bones to grow. Ask of each how fast it works. Nerves are fast, and they stop straight away. Growing takes years. So look for a slow message. And it happened while he slept.

ASK 14W

Theo grew four centimetres, mostly while asleep. What told his bones to do that?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

your blood carries things all around your body

VERDICT 70W

The body has two ways to send messages. Nerves are fast, like pulling your hand off something hot. Hormones are slow. They are chemicals that travel in the blood and can take hours or years to finish their job. Growth hormone is one of them, and the body makes most of it during sleep. That is why Theo grew while he was asleep, and why sleep is part of growing.

TAKEAWAY 15W

The body sends slow chemical messages, and some of them tell the body to grow.

4.2 Touch, Think, Move

BRAIN & SENSES · NEUROLOGY & SENSES · A PERSON · SEQUENCE

SCENE 57W

Lily touches something very cold. She pulls her hand back fast. The brain doctor shows four cards. It felt like one instant to Lily. It was really a sequence. Something was noticed. A message travelled. The brain understood it. A muscle moved. Each stage takes a tiny bit of time. So the order can be worked out.

GUIDE 40W

These four are one chain. Ask what has to happen first. You cannot answer something you have not noticed. So the skin comes before the thinking. And the moving comes after both. It feels like one thing. It is four.

ASK 5W

Put the stages in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

you pull your hand back from something cold

VERDICT 53W

You cannot answer something you have not noticed yet. So the skin senses the cold first. Nerves carry that message inward. The nervous system works out what it means. Then a message goes back out to the muscles and the hand moves. It feels like one thing. It is really four, in order.

TAKEAWAY 10W

The nervous system helps the body notice changes and respond.

4.3 After the Head Bump

BRAIN & SENSES · NEUROLOGY & SENSES · A PERSON · TRIAGE

SCENE 63W

Three children come in after playground falls. Arjun bumped his head. Now he seems confused. He has trouble answering simple questions. Bea has a small scrape on her elbow. Cole has a sore finger and talks normally. Clues about how someone thinks and answers matter a lot. The brain is hard to check from outside. Always tell an adult after a head bump.

GUIDE 40W

All three need seeing. Ask of each what could be getting worse right now. You can see a scrape. You cannot see a brain. So a change in how somebody thinks is the clue that matters after a head bump.

ASK 7W

Who should the hospital team check first?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

your brain is inside your head

VERDICT 57W

You can see a scrape and you cannot see a brain. So a change in how somebody thinks is the clue that matters after a head bump. Arjun is mixed up, and that is new. Bea's scrape can be cleaned. Cole's finger can wait for a picture. Only one of these could be getting worse right now.

TAKEAWAY 19W

In real life, children should tell an adult right away after a significant head injury or new concerning symptoms.

4.4 Observation or Measurement? — Review 2

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A ROOM · CALLBACK · CHOICE

SCENE 34W

Luis says his belly hurts. You can see that he is awake and talking. A nurse checks his temperature and pulse. The chart now contains both things the team noticed and numbers from tools.

GUIDE 41W

All four clues are real and useful. They come from different places. One you see. One the patient tells you. One comes from a tool, as a number. Ask of each where it came from. That is what decides this one.

ASK 5W

Which item is a measurement?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a tool can give you a number

VERDICT 56W

Clues come in three kinds. Some you see, like a worried face. Some the patient tells you, like where it hurts. Some come from a tool, as a number. A thermometer gives a number. That makes it a measurement. All three kinds are useful. Knowing which kind you have tells you how sure you can be.

TAKEAWAY 13W

Doctors combine observations, patient stories, and measurements. They are different kinds of clues.

END OF THE SHIFT 2W

Shift complete

Heart on the Move

PLAN CARD 80W

Fifth shift. Jordan came up the stairs and his heart is racing, and his mum looks frightened. Jordan says he feels fine and wants to go back to his brother. His hands are warm and he is not out of breath now. Nurse Alex Lee has counted his pulse twice, ten minutes apart, and asks you what you saw before she says anything. Today you follow blood around the body, count a pulse, and work out why it sped up.

BRIEFING 6W

The heart, blood, pulse, and circulation

5.1 The Blood Loop

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A ROOM · SEQUENCE

SCENE 27W

A heart model sits on the table. Four arrow cards lie beside it. The nurse says blood moves in a loop. Heart, lungs, heart, body, and back.

GUIDE 40W

These four make a loop. Ask what each step needs before it. Blood cannot bring oxygen to the body before it picks any up. And it picks it up in the lungs. Start at the heart and follow it round.

ASK 11W

Put the simplified blood loop in order, starting at the heart.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

the heart is a pump

VERDICT 52W

Blood does not go one way and stop. It goes around and around. The heart pumps it to the lungs. In the lungs it picks up oxygen. The heart pumps it out to the body. Then it comes back. That is a loop, and it never stops all day or all night.

TAKEAWAY 8W

Circulation is a loop, not a one-way trip.

5.2 Pulse Math

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A PERSON · BALLPARK

SCENE 27W

Jordan is resting after the stairs. The therapist counts 18 beats in 15 seconds. She asks you for the one minute number. This is math practice only.

GUIDE 42W

Two things matter here. How many beats, and how long you counted for. Ask what the question wants. It asks about a whole minute. So work out how many of your parts fit in one. Nurses count short blocks to save time.

ASK 13W

18 beats in 15 seconds is about how many beats in 60 seconds?

TAKES AS READ

a minute has 60 seconds

VERDICT 46W

1 minute has 4 15-second parts. So 18 beats in 15 seconds is 18 4 times, which is 72. Each beat is one push of blood. You can feel that push with two fingers and a clock. It is one of the oldest checks in medicine.

TAKEAWAY 13W

A pulse is a clue that the heart is beating and moving blood.

5.3 Why the Faster Pulse?

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A PERSON · VERIFY

SCENE 26W

Jordan's pulse was slow while he sat. It was fast right after the stairs. He feels fine now. The doctor asks you why it went up.

GUIDE 41W

Say what his pulse will be right after the stairs. Write your number down first. Then he climbs, and you count his pulse for a minute. Sitting still it was 76. Working muscles need more oxygen, and the blood carries it.

ASK 13W

Say what Jordan's pulse will be right after the stairs. Then count it.

BEHIND THE BUTTON 97W

What a pulse is. Each beat pushes blood along, and you can feel it in the wrist. Counting the beats in a minute tells you how fast the heart is working.

Why muscles change it. Muscles need oxygen to work. Blood carries oxygen. When the muscles work harder, the heart beats faster to send the blood round faster.

Why guessing first matters. If you look at the number and then explain it, any answer sounds right. If you say the number first, the count can tell you whether you understood what the body was going to do.

TAKES AS READ

muscles work harder when you climb stairs

VERDICT 84W

Jordan sat still first. His pulse was 76. Then he climbed the stairs. His leg muscles worked hard. Working muscles need more oxygen. Oxygen rides in the blood. So his heart beat faster to move the blood along. His pulse went up to about 120. He rested, and it came back down. Four body parts did one job. Muscles asked. Lungs took in air. Blood carried it. The heart pushed the blood. Nothing is wrong with Jordan. That is what a body should do.

TAKEAWAY 13W

Body systems work together. Muscles, lungs, blood, and heart all change during exercise.

5.4 Your First Exam Room — Review 2

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A ROOM · CALLBACK · SEQUENCE

SCENE 29W

Ben is here for his ankle check. The team has four things to do. Clean hands. Ben's story. A look at the ankle. A note for the next person.

GUIDE 38W

All four steps happen. So ask what each one needs first. What is on your hands goes on to Ben. His story tells you where to look. And the note is for the nurse who comes after you.

ASK 7W

Put these steps in the best order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

germs can travel on hands

VERDICT 44W

Each step makes the next one work. Hands come first. What is on them goes on to Ben. His story comes next. It tells you where to look. Then you look. Then you write it down, so the next person knows what you saw.

TAKEAWAY 14W

Each step in a hospital routine is there to make the next one work.

END OF THE SHIFT 2W

Shift complete

Sense Detective

BEFORE THE SHIFT — HUNT

Six pumps, and the list says four

Each pump helps one child get their medicine. Two are not where the list says they are. A pump nobody can find is no help at all. Go and find them before somebody needs one.

PLAN CARD 77W

Sixth shift. Mina says the board looks blurry, and she can read the big letters but not the small ones. Caleb cannot hear a quiet tone, though he hears you fine when you face him. Mrs. Ruiz feels dizzy when she stands, and is steady again after a minute. Three people, three clues, three different tools, and Nurse Alex Lee has put them all in one room. Today you match each one and follow how seeing works.

BRIEFING 9W

Eyes, ears, senses, and how the brain uses signals

6.1 Which Test Fits the Clue?

BRAIN & SENSES · NEUROLOGY & SENSES · A ROOM · CHOICE

SCENE 61W

Mina says the board looks blurry from far away. Her book looks fine. The eye doctor wants to start simple. A test is really a question. A good first test asks what the patient just asked. Here that is what Mina can see at different sizes. Starting with the matching test saves time. It also points at what to do next.

GUIDE 37W

Four tools, all of them good. Ask of each what it measures. Eyes? Ears? Sounds inside? Bones? Then look at the clue and match it. Every one of these answers some question. Only one answers this one.

ASK 6W

Which tool best matches Mina's clue?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

different tools check different things

VERDICT 52W

Match the tool to the clue. The clue here is about seeing, so you want the tool made for eyes, and that is the eye chart. A hearing test checks ears. A stethoscope hears sounds. An X-ray shows bones. Every one of them is a good tool. Only one answers this question.

TAKEAWAY 12W

The best test depends on the question you are trying to answer.

6.2 Match the Sense

BRAIN & SENSES · NEUROLOGY & SENSES · A PERSON · CHOICE

SCENE 55W

The team has three clue cards. Mina has blurry vision. Caleb has trouble hearing a tone. Mrs. Patel feels unsteady when she turns her head. Each sense starts with a part built to catch one kind of signal. Light. Sound. Movement. Here is the surprise. One organ can hold more than one of those jobs.

GUIDE 36W

Four sets, and one body part does two jobs. Ask of each person what they cannot do. Seeing? Hearing? Staying steady? The ear takes in sound, and it also tells the brain which way is up.

ASK 5W

Which body-part match is best?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

your eyes see and your ears hear

VERDICT 54W

Eyes take in light, and that is how we see. Ears take in sound, and that is how we hear. But the inner ear does a second job too. It tells the brain which way is up. That is why a problem in there can make somebody dizzy even when their hearing is fine.

TAKEAWAY 16W

One organ can have more than one job; the ear helps with both hearing and balance.

6.3 From Light to Picture

BRAIN & SENSES · NEUROLOGY & SENSES · A PERSON · SEQUENCE

SCENE 59W

Jules wants to know how light becomes a picture. The brain doctor lays out four cards. Eyes do not really see on their own. They catch light and turn it into signals. The picture is put together later, in the brain. That is why seeing is a team job. Knowing the stages tells you where a problem could be.

GUIDE 41W

These four are one path. Ask what has to happen first. The brain cannot make sense of a pattern the eye has not picked up. So the eye collects, and the brain understands. A problem could be anywhere along the line.

ASK 7W

Put the simplified vision pathway in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

you need light to see

VERDICT 51W

Seeing is not one step. Light comes into the eye. The back of the eye picks up the pattern. A nerve carries that to the brain. The brain works out what it is. So the eye collects, and the brain understands. A problem could be at any point along that line.

TAKEAWAY 11W

Seeing is a team job between the eyes, nerves, and brain.

6.4 Follow the Sandwich — Review 2

FOOD & WATER · NUTRITION & KIDNEY CARE · A ROOM · CALLBACK · SEQUENCE

SCENE 31W

Owen has a paper sandwich card. He wants to know where it goes after he swallows. The dietitian hands you four organ cards and asks you to put them in order.

GUIDE 40W

These four are one journey, and each part has one job. Ask what has to happen first. Nothing can soak up tiny bits before something has made them tiny. Food is far too big for the body to use whole.

ASK 6W

Put the food journey in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

food gets chewed before you swallow it

VERDICT 57W

Food is far too big for the body to use. It gets broken down in stages. The mouth chews it and wets it. It goes down the food tube. The stomach mixes it and breaks it down more. Then the small intestine takes the tiny bits into the blood. Each part does one piece of the job.

TAKEAWAY 9W

Different organs have different jobs in one connected system.

END OF THE SHIFT 2W

Shift complete

Breathing Room

PLAN CARD 73W

Your seventh shift. Noah helped clean a dusty room and now he makes a whistling sound with every breath. He is sitting up very straight and would rather not lie down. Dr. Luis Reyes wants you to listen before he does. He has already checked that Noah can still talk in whole sentences. Today you follow the path air takes, work out what that sound means, and count how fast someone is breathing.

BRIEFING 6W

Lungs, air, oxygen, and breathing rate

7.1 Follow One Breath

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A ROOM · CHAIN

SCENE 30W

Priya takes a deep breath. She asks where the air goes. She is also wheezing a little today, and the team wants to know where the air is having trouble.

GUIDE 47W

Put the four parts of the path in the order the air travels. Then name the part where the air is having trouble today. Each part has a number beside it, saying how much air it can let through. The biggest part is not always the problem.

ASK 16W

Put the breath's path in order, then name the part where the air is having trouble.

BEHIND THE BUTTON 98W

What the path is. Air goes in through the nose and mouth, down the windpipe, into smaller and smaller tubes, and reaches tiny air sacs. The blood picks up oxygen there.

What a wheeze is. A whistling sound air makes when it squeezes through a narrow tube. That tells you where to look: the small tubes, not the nose and not the sacs.

Why the narrow part decides. Air can only get through as fast as the tightest part of the path lets it. Opening up the nose more would change nothing while the small tubes are squeezed.

TAKES AS READ

air goes in through your nose and mouth

VERDICT 105W

A breath goes in the nose and mouth. Then down the windpipe. Then into the small tubes. Then into the air sacs. The blood picks up oxygen there. Today one part is not working well. Look at the numbers. The nose is fine. The windpipe is fine. The air sacs could take much more air. The small tubes let through less than half. That is where the wheeze comes from. Air whistles when it squeezes through a narrow tube. The tightest part decides how much air gets through. So helping the small tubes open up will help Priya. Breathing harder through her nose will not.

TAKEAWAY 12W

The lungs are the meeting place between outside air and the blood.

7.2 The Dusty-Room Clue

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A PERSON · CHOICE

SCENE 60W

Noah helped clean a dusty storage room. Now he is wheezing. He can still talk. Each breath makes a small whistling sound. The team is with him. One clue by itself does not mean much. The useful question is which body part could make that clue. A whistle needs air squeezing through a narrow space. So the clue points somewhere.

GUIDE 43W

Four body parts, and one sound. Ask of each whether it could whistle on every breath. A whistle needs air squeezing through a narrow gap. So ask what the air is passing through. One clue alone means little. This one names a part.

ASK 9W

Which body system should the team focus on first?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

blowing through a small gap makes a whistle

VERDICT 52W

A whistle needs air squeezing through a narrow space. Noah makes that sound with every breath. So it is the breathing tubes that got narrow. That is what wheezing is. One clue on its own does not mean much. This one is useful because you can name the part that made it.

TAKEAWAY 16W

A clue is useful when you connect it to the body system that could produce it.

7.3 Count the Breaths

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A PERSON · BALLPARK

SCENE 25W

Mrs. Grant is resting. The nurse counts 5 breaths in 15 seconds. She asks you for the one minute number. This is math practice only.

GUIDE 41W

Two things matter here. How many breaths, and how long you counted for. Ask what the question wants. It asks about a whole minute. You counted part of one. So work out how many of your parts fit in a minute.

ASK 13W

5 breaths in 15 seconds is about how many breaths in 60 seconds?

TAKES AS READ

a minute has 60 seconds

VERDICT 45W

1 minute has 4 15-second parts. So 5 breaths in 15 seconds is 5 4 times, which is 20. Nurses count short blocks to save time. People also breathe differently when they know somebody is watching. So a short count is often the honest one.

TAKEAWAY 8W

Doctors often turn short counts into one-minute estimates.

7.4 Why Food Is Fuel

FOOD & WATER · NUTRITION & KIDNEY CARE · A ROOM · CALLBACK · CHOICE

SCENE 26W

Nina ran all morning and now she feels shaky. She has not eaten since breakfast. Her nurse asks you what her body has run out of.

GUIDE 39W

Four reasons Nina might feel shaky. Ask of each whether the story fits. She ran all morning. She has not eaten. Two of these answers are about water or about nerves. Food is fuel, and running spends it fast.

ASK 13W

Nina ran all morning and has not eaten. Why does she feel shaky?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

food gives your body energy

VERDICT 63W

Food is fuel. Your body breaks food down and uses the energy to move, to keep warm, and to think. Running uses a lot of energy fast. Nina ate at breakfast and then used it all up. When the fuel runs low, people feel shaky, tired, or grumpy. A snack puts the fuel back. This is why hospitals ask when someone last ate.

TAKEAWAY 15W

The body spends what it takes in, and spends it faster when it works harder.

END OF THE SHIFT 2W

Shift complete

Water Watch

PLAN CARD 74W

Eighth shift. Mateo finished his soccer match and says he feels fine. His mouth is dry. His urine looks darker than usual. He drank one small bottle in the whole match. Nurse Alex Lee has written all of that down and wants to know what you make of it together. Today you read two clues at once and follow water through the body. Two clues pointing the same way are much stronger than one.

BRIEFING 6W

Hydration, kidneys, urine, and water balance

8.1 The Thirsty Player

FOOD & WATER · NUTRITION & KIDNEY CARE · A ROOM · BALLPARK

SCENE 27W

Mateo just finished soccer. He is thirsty and his mouth is dry. The nurse says his urine looks darker than it was. The team is checking him.

GUIDE 46W

Work out how much water Mateo lost while he played. He sweated about half a litre every twenty minutes, and he played for sixty. Pick the two numbers that go together. One tile is how long a half is, which is not what you need here.

ASK 10W

How much water did Mateo lose while he was playing?

BEHIND THE BUTTON 100W

Why you sweat. Running makes heat. Sweat carries the heat away when it dries off your skin. That works well, and it uses up water.

What the clues mean. A dry mouth and dark urine both say the same thing: there is less water in the body than there should be. The body saves water by making less urine, and that makes it darker.

What to do about it. Drink. A litre and a half is about three big glasses, and you cannot drink it all at once, which is why players drink through a game instead of after it.

TAKES AS READ

you sweat when you run around

VERDICT 108W

Mateo played for 60 minutes. He sweated about half a litre every 20 minutes. Sixty minutes holds three lots of twenty. So he lost about $3 \times 0.5 = 1.5$ litres. That is about three big glasses. It went into the air off his skin. Sweating is not a fault. It carries heat away. Without it he would get too hot. But the water has to go back in. His dry mouth is one clue. His dark urine is another. The body saves water by making less urine. So the urine looks darker. Both clues say the same thing. He needs a drink, a bit at a time.

TAKEAWAY 17W

The body tells you when it is short of something, and thirst is one of those signals.

8.2 The Drink Tracker

FOOD & WATER · NUTRITION & KIDNEY CARE · A PERSON · BALLPARK

SCENE 46W

Mrs. Lee has a drink chart by her bed. It shows 2 cups in the morning, 1 cup at lunch, and 3 cups later. A wall chart says a body her size loses about 8 cups a day. This is a math problem, not drink advice.

GUIDE 39W

Two amounts matter here. What went in, and what went out. Ask what the question wants. It asks how far behind she is. That is a difference, so you need both. Add the drinks up before you compare them.

ASK 11W

How many cups short of what her body lost is she?

TAKES AS READ

you can add up a list of numbers

VERDICT 46W

Six cups went in. About eight cups went out. That leaves her two cups short. The body loses water all day, even when you are still. So you count what goes in against what goes out. One number on its own does not tell you much.

TAKEAWAY 15W

Water balance is what goes in measured against what goes out, not either one alone.

8.3 What the Kidneys Do

FOOD & WATER · NUTRITION & KIDNEY CARE · A PERSON · BALLPARK

SCENE 33W

Harper looks at a model. It shows the kidneys and the bladder. She asks why both are in one system. The body makes waste. The blood carries it. The kidneys are the filters.

GUIDE 49W

Work out how much water the kidneys send back into the body each day. They filter 180 litres, and only 2 litres leave as urine. Take the smaller number away from the bigger one. One tile is how many kidneys there are, which is not part of this sum.

ASK 14W

How much water do the kidneys put back into the body in a day?

BEHIND THE BUTTON 110W

What filtering means. The blood goes through the kidneys again and again. Each time, water and waste are pulled out, and then almost all of the water is put back. Only the waste, and a little water, carries on to the bladder.

Why the number is so big. The kidneys do not filter the blood once. They filter it all day, so the amount they handle is far bigger than the amount anybody drinks.

Why the bladder is in the same system. The kidneys make the urine, a little at a time, all day. Nobody could visit a toilet that often, so the bladder holds it until it is convenient.

TAKES AS READ

your body makes waste it has to get rid of

VERDICT 108W

The kidneys filter about 180 litres a day. Only about 2 litres leave as urine. So they put back $180 - 2 = 178$ litres. Filtering is not the same as throwing away. Nearly all of it goes back in. What carries on is the waste. It takes a little water with it. Nobody drinks 180 litres. The number is that big because the blood goes round and round. The kidneys clean it all day. Then the bladder. The kidneys make urine slowly, all day. Nobody could go to the toilet that often. So the bladder holds it. One system, two jobs. The kidneys filter. The bladder stores.

TAKEAWAY 5W

Kidneys filter; the bladder stores.

8.4 Why Take an X-Ray? — Review 2

BONES, MUSCLES & SKIN · IMAGING & REHAB · A ROOM · CALLBACK · CHOICE

SCENE 30W

Eli fell from his bike. His wrist hurts and is swollen. The doctor wants a picture that can help the team look for a broken bone without opening the skin.

GUIDE 38W

Four tools, and all four are good ones. Ask of each what it shows. Warmth? Sound? A shape? A size? The clue here is about a bone under the skin. So which tool shows what you cannot see?

ASK 7W

Which hospital tool best matches that job?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

some things are hidden under the skin

VERDICT 50W

You cannot see a bone by looking at an arm. An X-ray makes a picture of what is under the skin, and bones show up clearly in it. That is why it is the right tool here. A thermometer gives warmth. A stethoscope gives sound. Neither one shows a shape.

TAKEAWAY 11W

Doctors choose tools based on the kind of clue they need.

END OF THE SHIFT 2W

Shift complete

Germ Detective Shift

BEFORE THE SHIFT — CANVASS

Ask the children how they slept

The night nurse wrote down that the ward was quiet. Some of the children say it was not. How well they slept helps the doctors decide what to change today. Ask enough of them to be sure.

PLAN CARD 76W

Ninth shift. Ty and Ana are both tired today. They are in the same class, so everybody has decided it is the same bug. Dr. Amaka Okafor has not. Ty has a fever. Ana does not. They sat at different tables all week. Ana's little brother was sick last weekend. Today you work out how germs travel, why washing hands matters, and whether these two really match. Saying not enough evidence yet is a real answer.

BRIEFING 5W

Microbes, spread, evidence, and handwashing

9.1 How Might It Spread?

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A ROOM · CHOICE

SCENE 58W

The nurse gives you three short stories. Ty was near classmates who were coughing. Ana ate food that made several relatives sick. Lena touched a shared toy and rubbed her nose. Germs cannot fly or walk by themselves. Something has to carry them. Different germs travel different ways. The way it travels tells you how to stop it.

GUIDE 36W

Four sets, and germs travel in more than one way. Ask of each person what they did. Coughed near others? Ate something? Touched a handle? The story is what tells you which way the germ came.

ASK 6W

Which matching set is most sensible?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

germs are too small to see

VERDICT 54W

Germs do not all travel the same way. Some ride in tiny drops from a cough. Some are in food that was not cooked or kept right. Some sit on a doorknob until a hand picks them up. That is why the story matters. What somebody did tells you which way the germ came.

TAKEAWAY 10W

Knowing the route helps people choose the right prevention strategy.

9.2 Handwashing Challenge

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · CONTROL

SCENE 35W

Lena asks why there are sinks everywhere. Hands touch more things than any other body part. So hands carry the most germs. The team has a glowing powder. It shows where hands are still dirty.

GUIDE 54W

Four things are different between a good wash and a quick rinse. Change one of them and wash again. The lamp counts the glowing spots left on your hands. Change two things at once and you will not know which one helped. Then put your one back and see if the spots come back.

ASK 13W

Which one thing makes the most difference to how clean your hands get?

BEHIND THE BUTTON 99W

What soap does. Germs and grease stick to skin. Soap lets water lift them off, so they run away down the drain instead of staying on your hands.

Why one thing at a time. If you use soap and scrub for longer and use warm water, and the spots go away, you still do not know which of the three did it. Change one, and the lamp tells you.

Why putting it back matters. If the spots come back when you stop using soap, then it was the soap. If they stay away, something else was doing the work.

TAKES AS READ

soap and water clean things

VERDICT 106W

Four things were different. Only one of them matters much. Wash with soap and keep the rest the same. The lamp finds about 4 spots instead of 34. Stop using soap, and the spots come back. That is how you know it was the soap. The others do much less on their own. Warm water hardly changes the count. Twenty seconds with no soap beats five seconds, but not by much. A paper towel matters for what you touch next. Germs and grease stick to skin. Soap lets the water lift them off. Then they run down the drain. That is why there are sinks everywhere.

TAKEAWAY 13W

Handwashing is a simple action that can break a chain of germ spread.

9.3 Same Symptom, Same Germ?

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · CHOICE

SCENE 64W

Ty and Ana both feel tired. The rest of their clues are different. So is where each of them had been. A student says they must have the same germ. The germ scientist asks you to judge that. Many different illnesses make people tired. One shared clue is weak proof. Medicine is a puzzle. The pattern across clues is what points to an answer.

GUIDE 39W

Four answers, and two of them say yes. Ask of each what it takes as proof. Lots of things make a person tired. So one shared feeling is not enough. Saying not enough evidence yet is a real answer.

ASK 5W

What is the best response?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

being tired can happen for many reasons

VERDICT 53W

One symptom can have many causes. Lots of things make a person tired. So two tired people do not have to share a germ. You need more clues before you can say. This is a hard habit, because guessing feels faster. Saying not enough evidence yet is a real answer, not a dodge.

TAKEAWAY 10W

Medicine is an evidence puzzle, not a one-clue guessing game.

9.4 Three Jobs, Three Organs

FOOD & WATER · NUTRITION & KIDNEY CARE · A ROOM · CALLBACK · CHOICE

SCENE 30W

Three cards each need one picture. Owen needs the part that starts breaking food up. Sam needs the part that mixes it. Nia needs the part that soaks it up.

GUIDE 38W

Four sets, and only one has every job in the right place. Ask what each part does. Chewing makes pieces smaller. Mixing breaks them down more. Soaking up comes last. Check all three before you pick a set.

ASK 5W

Which matching set is correct?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

different body parts do different jobs

VERDICT 47W

A body system works like a good team. Nobody does everything. The mouth chews, so the pieces get smaller. The stomach mixes and breaks them down more. The small intestine soaks up the tiny bits. Each part is built for its own job, and the order matters.

TAKEAWAY 14W

A body system works because specialized parts each do a piece of the job.

END OF THE SHIFT 2W

Shift complete

Immune Team

PLAN CARD 73W

Tenth shift. Ruby is getting better. Nobody gave her anything, and that puzzles her. She had this illness two years ago and does not remember it. Mae is here for a shot. She is frightened of it. She wants to know how long it will hurt for. Today you follow what a body does when a germ gets in, and learn what a shot is for. A shot lets the body practise first.

BRIEFING 8W

Barriers, white blood cells, antibodies, and immune memory

10.1 The Body Notices a Germ

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A ROOM · SEQUENCE

SCENE 57W

Ruby is getting better. The immune doctor wants to show why. Four picture cards show what her body did. The steps happen in order. Something has to be noticed before it can be fought. Helpers have to arrive before the cleanup. The last step is the surprising one. The body keeps a record of what it met.

GUIDE 39W

These four are one order. Ask what has to happen first. Nothing gets attacked before it is spotted. And nothing is remembered before the fight. That last step is why you often do not get the same illness twice.

ASK 7W

Put the body's germ-fighting steps in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

your body can fight off a cold

VERDICT 53W

The immune system does four things, in order. A germ gets past the outside walls. Immune cells spot that something does not belong. They attack it and clear it out. Then some of them remember what it looked like. That last step is why you often do not get the same illness twice.

TAKEAWAY 15W

Immune memory can help the body respond faster when it meets the same threat again.

10.2 Meet the Defense Team

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · CHOICE

SCENE 62W

Three cards need labels. Dante has a skin shield card. Ruby has a white blood cell card. Mae has an antibody card. The body does not defend itself one way. It defends in layers. Barriers try to keep germs out. Cells patrol and attack what gets in. Special molecules know one threat very well. Each layer catches what the last one missed.

GUIDE 38W

Four sets, and each part has one job. Ask what each one does. Keeps things out? Goes looking? Knows one target? None of them can do the other two jobs. Check all three before you pick a set.

ASK 6W

Which set is the best match?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a team works better than one person alone

VERDICT 54W

Your defences are a team, and each part has one job. Skin is the wall that keeps things out. White blood cells go and find things that got in. Antibodies are the part that knows one target and sticks to it. They work together, and none of them can do the other two jobs.

TAKEAWAY 9W

Defense works in layers: barriers, cells, and specialized molecules.

10.3 A Vaccine Visit

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · SEQUENCE

SCENE 30W

Mae is at the clinic with her mum. A nurse will give the shot. A shot lets the body practise, so it is ready if the real germ turns up.

GUIDE 37W

All four steps happen. Ask which one cannot be taken back. A shot cannot, so checking who it is comes first. Then telling Mae, so she can hold still. The wait at the end is not politeness.

ASK 8W

Put the clinic process in the best order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a shot cannot be taken back once it is given

VERDICT 56W

Check who it is first. A shot cannot be taken back. Then tell Mae what is coming, so she can hold still. Then the nurse gives it. Then you wait a few minutes. That wait is not just being polite. If a rare fast problem starts, it starts right then, and a nurse is still there.

TAKEAWAY 23W

The wait at the end of a vaccine visit is part of the visit: fast reactions show up in the first few minutes.

10.4 Skeleton Jobs — Review 2

BONES, MUSCLES & SKIN · IMAGING & REHAB · A ROOM · CALLBACK · CHOICE

SCENE 36W

The orthopedic doctor has three teaching cards. Rosa needs the part that protects the heart and lungs, Mr. Bell needs the structure that supports the back, and Tess needs the place where bones meet and move.

GUIDE 38W

Four sets, and bones do more than one job. Ask what each bone is shaped for. A cage guards what is inside it. A column holds you up. A joint is where two bones meet, so it bends.

ASK 4W

Which set is correct?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

bones are hard and do not bend

VERDICT 55W

Bones do more than one job. The rib cage is a cage, so it guards the heart and lungs inside it. The spine holds you up and guards the nerve cord running down your back. A joint is a place where two bones meet, and that is where you bend. Same material, three different jobs.

TAKEAWAY 9W

The skeleton supports, protects, and helps the body move.

END OF THE SHIFT 2W

Shift complete

Rehab Race

PLAN CARD 76W

Eleventh shift. Sophie has been doing her hand exercises for a week and her fingers still will not close all the way. She wants to stop. She can close them nearly all the way in the morning and less by the afternoon. Her mum has been counting the sets at home and has the list with her. Today you follow how a body moves, learn why muscles get tired, and count out a set of exercises.

BRIEFING 5W

Muscles, joints, movement, and practice

11.1 Make the Hand Move

BONES, MUSCLES & SKIN · IMAGING & REHAB · A ROOM · SEQUENCE

SCENE 62W

Sophie wiggles her fingers. She asks how the message got there so fast. The therapist gives you four cards. Moving one finger takes a whole chain. First a decision. Then a message along a nerve. Then a muscle pulls. Then a bone moves at a joint. Therapists think about this chain a lot. When movement fails, the chain shows where to look.

GUIDE 38W

These four are one chain. Ask what has to happen first. A muscle cannot pull before the message reaches it. And bones do not move themselves. Muscles only pull, never push, which is why they come in pairs.

ASK 6W

Put the movement pathway in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

your brain is in charge of your body

VERDICT 47W

Moving takes four things in order. The brain decides. Nerves carry that message out to the muscles. The muscles tighten and pull. Bones move at the joints. Muscles can only pull, never push, which is why they come in pairs. Nothing moves until the message gets there.

TAKEAWAY 11W

One movement needs several parts of the body working in turn.

11.2 Tired Muscles

BONES, MUSCLES & SKIN · IMAGING & REHAB · A PERSON · CONTROL

SCENE 31W

Amara has done the same gentle motion many times. Her arm feels tired now. Nothing hurts sharply. There is no new injury. The therapist stops the set and lets her rest.

GUIDE 60W

Four things are different about the sets where her arm got tired quickly. Change one and run the set again. The grip meter shows how hard she can still squeeze at the end. Change two at once and the meter cannot tell you which one mattered. Then put your one back and see if the tiredness comes back with it.

ASK 9W

Which one thing decides how tired Amara's arm gets?

BEHIND THE BUTTON 118W

Why muscles get tired. Every pull uses energy. Do it enough times without a break and the muscle cannot pull as hard, until it has had time to recover.

Why tired is not hurt. Tiredness comes on slowly, feels the same all over the muscle, and goes away with rest. A new injury usually hurts sharply in one spot and does not get better while you rest for a minute.

Why the test matters. If rest is what fixes it, the answer is more rests. If something else is doing it — too much weight, too fast, too many in a row — then the plan changes, and only one test at a time can tell you which.

TAKES AS READ

you get tired after doing something over and over

VERDICT 125W

Four things were different between the sets. Only one changes how tired her arm gets. Give her a rest every ten pulls. Keep the weight, the speed and the movement the same. Her grip at the end goes from 12 up to 26. Take the rest away, and it drops back to 13. So it was the rests. The meter follows the change both ways. The others do far less on their own. More weight, a faster movement, a different chair — each one moves the meter less than its own wobble. Muscles use energy every time they pull. They need time to get it back. That is why the therapist stopped. Tired comes on slowly and goes away with a rest. Hurt does not.

TAKEAWAY 13W

Body tissues can change their activity depending on what the body is doing.

11.3 Therapy Sets

BONES, MUSCLES & SKIN · IMAGING & REHAB · A PERSON · BALLPARK

SCENE 58W

Mr. Diaz is doing gentle leg lifts. His therapist picked 3 sets of 5. The therapist chose the plan, not you. Therapy uses sets because short rests help muscles. Counting in groups is also easier to track. A nurse can count to five while talking. 15 is harder. Groups of the same size are what multiplying is for.

GUIDE 35W

Two things matter here. How many sets, and how many lifts in each. Ask what the question wants. It wants the whole lot. So one of them has to happen the other number of times.

ASK 10W

How many leg lifts are in 3 sets of 5?

TAKES AS READ

a set is a group of the same thing

VERDICT 47W

There are 3 sets, and each set has 5 lifts. So it is 5 3 times, which is 15. This is why therapy is counted in sets. Counting the total helps the team see if it is getting easier or harder from one day to the next.

TAKEAWAY 10W

Repeated sets are groups, so multiplication helps count them quickly.

11.4 The Body's Slow Messages

BRAIN & SENSES · NEUROLOGY & SENSES · A ROOM · CALLBACK · CHOICE

SCENE 30W

Theo is 11 and has grown 4 centimetres since spring. His chart shows he grew most while he was asleep. He wants to know who told his bones to grow.

GUIDE 40W

Four things that might have told his bones to grow. Ask of each how fast it works. Nerves are fast, and they stop straight away. Growing takes years. So look for a slow message. And it happened while he slept.

ASK 14W

Theo grew four centimetres, mostly while asleep. What told his bones to do that?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

your blood carries things all around your body

VERDICT 70W

The body has two ways to send messages. Nerves are fast, like pulling your hand off something hot. Hormones are slow. They are chemicals that travel in the blood and can take hours or years to finish their job. Growth hormone is one of them, and the body makes most of it during sleep. That is why Theo grew while he was asleep, and why sleep is part of growing.

TAKEAWAY 15W

The body sends slow chemical messages, and some of them tell the body to grow.

END OF THE SHIFT 2W

Shift complete

Skin Shield

BEFORE THE SHIFT — EVADE

Keep out of the way of the fast trolley

A child is being taken to the scanner right now. The hall is narrow. If you stand in it, they have to stop. Get where you are going the other way round, and let them past.

PLAN CARD 70W

Twelfth shift. Ella came off her bike. There is gravel in her knee. It has stopped bleeding on its own. June was outside too long and has a sunburn. It is on the tops of her shoulders and nowhere else. Today you follow how a scrape heals, work out what job the skin lost, and help without doing a grown-up's job. The body shuts the door first and repairs after.

BRIEFING 7W

Skin as a barrier, healing, and protection

12.1 How a Scrape Heals

BONES, MUSCLES & SKIN · IMAGING & REHAB · A ROOM · SEQUENCE

SCENE 57W

Ella has a small scrape. The team cleaned it and covered it. She asks what her body will do next. Skin is not a wrapper. It is living tissue. It fixes itself in stages, in the same order every time. First the bleeding stops. Then the cleanup. Then rebuilding. Knowing the stages tells you what is normal.

GUIDE 41W

These four are stages, not choices. Ask what has to happen first. New skin cannot grow under a lid that is not there. And a clot has to close the gap first. The body shuts the door, then does the repairs.

ASK 7W

Put the steps of healing in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a cut stops bleeding on its own after a while

VERDICT 54W

Healing goes in stages. First the skin is broken open. Then the blood clots and closes the gap. Then a scab forms and covers it like a lid. Under that lid, new skin grows. The body shuts the door first and does the repairs after. That is why picking a scab sets it back.

TAKEAWAY 10W

Skin is living tissue that can repair many small injuries.

12.2 Clean Team, Not DIY Doctor

BONES, MUSCLES & SKIN · IMAGING & REHAB · A PERSON · SEQUENCE

SCENE 26W

You are the junior doctor in this game. Real wound care is a job for grown-ups. Your job is to keep things safe and in order.

GUIDE 37W

All four steps happen. Ask which one takes time to arrive. Help does, so telling somebody comes early. Then your hands, because a cut is a way in. The grown-up does the caring. You keep the notes.

ASK 6W

Put the junior-doctor actions in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

some jobs are for grown-ups

VERDICT 49W

Tell a grown-up first. Help takes time to come, and nothing else has to wait for it. Then clean your hands. A cut is a way in, and hands are how germs move. Then get the things they need. The grown-up cares for the cut. You keep the notes.

TAKEAWAY 17W

A scrape is a break in the barrier, so everything that touches it has to be clean.

12.3 What Job Did the Skin Lose?

BONES, MUSCLES & SKIN · IMAGING & REHAB · A PERSON · CHOICE

SCENE 57W

June was outside too long. Now she has a mild sunburn. The skin doctor explains what skin does. It keeps water in. It keeps germs out. It shields everything under it. Sunlight also carries a kind of light we cannot see. Too much of it can hurt skin cells. So the shield needs shade and sunscreen too.

GUIDE 36W

Four jobs skin does, and it does all of them. Ask of each whether the scrape stopped it. Skin still feels touch. It still sweats. What a hole in it takes away is being a wall.

ASK 10W

Which main skin job is most connected to this lesson?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

skin covers the whole outside of you

VERDICT 50W

Skin does several jobs at once. It holds water in. It feels touch. It cools you with sweat. But its biggest job is being a wall. Nothing gets in through skin that is whole. A scrape makes a hole in the wall, and that is the job it just lost.

TAKEAWAY 11W

Skin protects the body, but the skin itself also needs protection.

12.4 Touch, Think, Move — Review 2

BRAIN & SENSES · NEUROLOGY & SENSES · A ROOM · CALLBACK · SEQUENCE

SCENE 17W

Lily touches something very cold and quickly moves her hand. The brain doctor shows four simplified stages.

GUIDE 40W

These four are one chain. Ask what has to happen first. You cannot answer something you have not noticed. So the skin comes before the thinking. And the moving comes after both. It feels like one thing. It is four.

ASK 5W

Put the stages in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

you pull your hand back from something cold

VERDICT 53W

You cannot answer something you have not noticed yet. So the skin senses the cold first. Nerves carry that message inward. The nervous system works out what it means. Then a message goes back out to the muscles and the hand moves. It feels like one thing. It is really four, in order.

TAKEAWAY 10W

The nervous system helps the body notice changes and respond.

END OF THE SHIFT 2W

Shift complete

Allergy Alert

PLAN CARD 76W

Thirteenth shift. This one moves fast. Chloe's lips are swelling and she says it is hard to breathe. She had a peanut biscuit ten minutes ago. Sofia's eyes have itched every spring for years. Nurse Alex Lee is already walking and wants your reason as you go. Today you decide who cannot wait. You tell an allergy from a germ. Then you hand over to the emergency team. Swelling with trouble breathing cannot wait for anything.

BRIEFING 8W

Allergies, immune overreaction, and recognizing urgent breathing clues

13.1 Which Allergy Patient First?

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A ROOM · TRIAGE

SCENE 62W

Three people come in with allergy signs. Sofia has itchy eyes and breathes normally. Malik has an itchy rash and talks normally. Chloe has swelling around her mouth. She is struggling to breathe. The team needs your call. An allergy is the body's defence going much too far. Most reactions stay about the same for a while. A few change very fast.

GUIDE 39W

All three need help. Ask of each what could get worse in minutes. Itchy eyes stay about the same. A big rash looks worse than a small one. How much skin it covers is not what makes it urgent.

ASK 7W

Who needs the team's immediate attention first?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

you cannot go long without breathing

VERDICT 54W

Swelling with trouble breathing can get dangerous in minutes. That is why Chloe goes first. Itchy eyes are horrible and they stay about the same. A big rash looks worse than a small one, but how much skin it covers is not what makes it urgent. You go by what could get worse fastest.

TAKEAWAY 7W

When breathing is threatened, urgency rises sharply.

13.2 Allergy or Germ Clues?

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · CHOICE

SCENE 58W

Sofia's eyes itch every spring. That is when pollen is high. She has no fever. Ty came in earlier with a fever and a cough. His classmates were sick too. The allergist asks which story sounds like an allergy. Both patients feel unwell. So no single symptom decides it. The pattern does. What else is happening, and when.

GUIDE 41W

Four answers, and both patients feel unwell. Ask of each what the pattern under it is. Does it come back every year with the pollen? Or did it start after being near somebody sick? Look at the story, not the feeling.

ASK 5W

Which comparison is most sensible?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

some things happen at the same time every year

VERDICT 48W

Look at the pattern, not just the word. Sofia itches when the pollen comes. It happens every year. That fits an allergy. Ty has a fever. He was near people who were sick. That fits a germ. Both feel unwell. The stories under them are not the same.

TAKEAWAY 12W

Doctors compare patterns, timing, and exposures instead of relying on one clue.

13.3 Emergency Team Handoff

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · SEQUENCE

SCENE 30W

The team is helping Chloe now. In this game you do not pick or give medicine. Your job is to get the right grown-ups fast and tell them the story.

GUIDE 40W

All four steps happen. Ask which clock is already running. The team takes time to come, so calling is first. Then watch her breathing, because that changes fastest. Writing it down is last, so nobody has to guess the order.

ASK 6W

Put the junior-doctor actions in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

help takes time to arrive

VERDICT 48W

Call first. That clock is running whether or not you do anything else. Then watch her breathing, because it can change fastest. Then say what she met and when. The team needs that before they act. Write it down last, so nobody has to guess the order later.

TAKEAWAY 24W

In an emergency the first action is the one with a clock on it, and the last is the one that protects the record.

13.4 Possible Broken Bone — Review 2

BONES, MUSCLES & SKIN · IMAGING & REHAB · A ROOM · CALLBACK · SEQUENCE

SCENE 27W

Eli's wrist may be hurt. He looks worried. In this game you do not fix it. You keep him safe and get the team what they need.

GUIDE 41W

All four steps happen. Ask which one stops things getting worse. Everything else can happen while the wrist is held still. Help takes time to come, so telling somebody is early. The note comes last, and it holds all of it.

ASK 7W

Put the safe hospital response in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

moving something that is hurt can make it worse

VERDICT 46W

Keeping the wrist still comes first. It is the only step that stops more harm. All the rest can happen while you hold it still. Then you tell a grown-up, because help takes time to come. The note goes last. It writes down all of it.

TAKEAWAY 18W

The first action is the one that stops things getting worse; the rest can happen while it holds.

END OF THE SHIFT 2W

Shift complete

Temperature Trouble

PLAN CARD 77W

Fourteenth shift. Jamal is hot and sweating after exercise. Ivy got caught in the rain and is shivering. Ms. Kim was 98 degrees at 9 o'clock and 101 at 12. Nurse Alex Lee wrote both of Ms. Kim's numbers down and left the times beside them. Today you work out what each body is doing and how fast the fever is climbing. One number is just a number. Two numbers and a clock tell you how fast.

BRIEFING 6W

Homeostasis, sweating, shivering, and body temperature

14.1 Hot or Cold Response?

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A ROOM · CHOICE

SCENE 60W

Jamal is hot after exercise. He is sweating. Ivy got caught in the rain. She is cold and shivering. Both are with the team now. The body works hard to hold one steady temperature. Cells only work well in a narrow range. When it gets too warm, the body loses heat. When it gets too cold, the body makes heat.

GUIDE 38W

Four answers about sweating and shivering. Ask of each whether the two do the same thing. Sweat dries, and drying takes heat away. Shivering is muscles working, and working muscles make heat. So one cools and one warms.

ASK 5W

Which explanation matches the clues?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

you sweat when you are hot and shiver when you are cold

VERDICT 52W

These two do opposite things. Sweat sits on your skin and dries. Drying takes heat away with it, so you cool down. Shivering is your muscles shaking fast, and working muscles make heat. So one cools and one warms. Your body uses both to stay at about the same temperature all day.

TAKEAWAY 13W

Homeostasis means the body works to keep important conditions within a useful range.

14.2 Temperature Difference

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · BALLPARK

SCENE 24W

A chart shows 2 numbers for Ms. Kim. The first is 98°F. The next is 101°F. The times beside them are three hours apart.

GUIDE 43W

Two things matter here. How much it changed, and how long that took. Ask what the question wants. It wants a speed, so it needs both. One number alone is just a number. The same change in ten minutes would worry you more.

ASK 10W

How fast is her temperature rising, in degrees per hour?

TAKES AS READ

you can take one number away from another

VERDICT 51W

101 take away 98 is 3. It took 3 hours. So it went up about 1 degree each hour. That is the whole trick. One number is just a number. Two numbers and a clock make a speed. The same 3 degrees in 10 minutes would be a much bigger worry.

TAKEAWAY 19W

A change on its own is not enough; how long it took is what says whether it is fast.

14.3 How Sweat Helps Cool

GERMS & DEFENSES · INFECTION & IMMUNOLOGY · A PERSON · VERIFY

SCENE 37W

Parker sees sweat on his arm. He has just been running. The nurse says water needs energy to dry up. It takes that energy from the warmest thing near it. On an arm, that is the arm.

GUIDE 43W

Say what his skin temperature will be five minutes after the fan goes on. Write your number down first. His skin is 34 degrees now, and it is damp. Drying water takes energy out of whatever is warmest, and that is his skin.

ASK 17W

Say what Parker's skin temperature will be five minutes after the fan goes on. Then measure it.

BEHIND THE BUTTON 108W

Why drying cools. Water needs energy to turn into vapour. It takes that energy from the nearest warm thing. When the warm thing is your skin, your skin gets cooler.

Why the fan helps. Moving air carries the vapour away, so more water can dry off. Still air above damp skin gets full of vapour and the drying slows down.

Why dry skin would not cool. If there is no sweat, there is nothing to dry, and the fan just moves air about. That is the difference between the sweat and the breeze — one of them is doing the cooling and the other is helping it along.

TAKES AS READ

water dries up in the air

VERDICT 108W

Water needs energy to dry up. It takes that energy from the warmest thing near it. On a damp arm, that is the arm. So the sweat dries and the skin gives up energy. The skin gets cooler. It went from 34 degrees to about 31 in five minutes. The fan is not doing the cooling. It just carries the damp air away. Then more sweat can dry. A fan on dry skin does almost nothing. There is nothing to dry. A wet cloth cools the same way on a hot day. This is why sweating uses up water. Somebody who has been sweating hard needs a drink.

TAKEAWAY 11W

The body uses physical processes, not magic, to help control temperature.

14.4 After the Head Bump — Review 2

BRAIN & SENSES · NEUROLOGY & SENSES · A ROOM · CALLBACK · TRIAGE

SCENE 35W

Three children come in from the playground. Arjun bumped his head. Now he looks muddled and cannot answer easy questions. Bea has a small elbow scrape. Cole has a sore finger but is talking normally.

GUIDE 40W

All three need seeing. Ask of each what could be getting worse right now. You can see a scrape. You cannot see a brain. So a change in how somebody thinks is the clue that matters after a head bump.

ASK 7W

Who should the hospital team check first?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

your brain is inside your head

VERDICT 57W

You can see a scrape and you cannot see a brain. So a change in how somebody thinks is the clue that matters after a head bump. Arjun is mixed up, and that is new. Bea's scrape can be cleaned. Cole's finger can wait for a picture. Only one of these could be getting worse right now.

TAKEAWAY 19W

In real life, children should tell an adult right away after a significant head injury or new concerning symptoms.

END OF THE SHIFT 2W

Shift complete

The Super Shift

BEFORE THE SHIFT — TAG

Catch Dr. Patel before she goes into the meeting

Dr. Patel runs the emergency department. She has the notes on Amara's breathing. The meeting she is going into lasts one hour. Nobody can plan Amara's day until they see those notes. Go and find her first.

PLAN CARD 81W

Your last shift. Six children are on the board and one room is free. Dr. Maya Patel, who runs the emergency department, is standing next to you and she is not going to say who goes first. Today you use all of it. Which clue means somebody can get worse quickly. Which tool answers which question. Which part of the hospital each child needs next. Then you hand over to the evening team, who were not here for any of today.

BRIEFING 8W

Putting body systems, evidence, triage, and teamwork together

15.1 Six Patients, One First Room

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A ROOM · TRIAGE

SCENE 58W

The waiting board is full. Nora breathes hard and cannot finish a sentence. DeShawn can walk on a sore ankle. Mei has a fever and cough but breathes fine. Mr. Alvarez feels lightheaded but is awake. Fatima has itchy eyes. Owen has a small covered scrape. One room is free. Triage asks which problem could get dangerous fastest.

GUIDE 39W

All six are real, and all six need something. Ask of each what happens while everybody waits. A sore ankle waits well. A fever waits. Feeling faint is worth a look. Only one of them can get worse quickly.

ASK 8W

Who should go to the first available room?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

only one person can go in first

VERDICT 50W

This is the big pattern of the whole course. Breathing trouble can get dangerous quickly, so it goes first. Feeling lightheaded is worth a look. A fever needs care. A sore ankle needs a picture. All of them are real. Only one of them is getting worse while everybody waits.

TAKEAWAY 9W

Triage means asking, “Which problem could become dangerous fastest?”

15.2 Send Them to the Right Place

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A PERSON · CHOICE

SCENE 58W

Nora is with the emergency team now. Five other patients still need a next stop. A hospital is not one big room. It is a set of special places. Each one is set up to answer a certain kind of question. The right place means the right test the first time. The wrong place costs a patient hours.

GUIDE 39W

Five people and five places. Pair them by asking what each clue needs. A picture of a bone? A check for germs? Some numbers taken? Send them all one place and most of them wait for the wrong thing.

ASK 5W

Which routing set is best?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a hospital has different rooms for different things

VERDICT 53W

Different clues need different tools. A sore ankle needs a picture of the bone. A fever needs a check for germs. Feeling faint needs some numbers taken. Itchy eyes need an allergy check. A scrape needs the wound team. Send them all one place and most of them wait for the wrong thing.

TAKEAWAY 9W

Hospitals are teams of specialized places that work together.

15.3 End-of-Day Handoff

TRIAGE & EMERGENCY · EMERGENCY & TRIAGE · A PERSON · SEQUENCE

SCENE 58W

Your shift is ending. The evening team is arriving. They were not here for any of today. They cannot read every note in time. A handoff is a summary built for them. Who each patient is. What matters most. What has been done. What to watch for. Careful work only helps if the next person can use it.

GUIDE 42W

All four things get said. Ask what the other nurse knows already. Nothing at all. So start with who it is and what is wrong. Then the clues, so they can judge. Get the order wrong and they hear the answer first.

ASK 8W

Put the handoff information in the best order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

the next person was not there when it happened

VERDICT 57W

A handoff is for somebody who missed everything. So it starts with who and what is wrong. Then the clues you have, so they can judge for themselves. Then what has already been done, so nobody repeats it. Then what still needs doing. Get the order wrong and they hear the answer before they know the question.

TAKEAWAY 15W

Science is only useful if the next person can understand and act on the evidence.

15.4 The Blood Loop — Review 2

BREATHING & HEART · RESPIRATORY & CARDIOLOGY · A ROOM · CALLBACK · SEQUENCE

SCENE 27W

A heart model sits on the table. Four arrow cards lie beside it. The nurse says blood moves in a loop. Heart, lungs, heart, body, and back.

GUIDE 40W

These four make a loop. Ask what each step needs before it. Blood cannot bring oxygen to the body before it picks any up. And it picks it up in the lungs. Start at the heart and follow it round.

ASK 11W

Put the simplified blood loop in order, starting at the heart.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

the heart is a pump

VERDICT 52W

Blood does not go one way and stop. It goes around and around. The heart pumps it to the lungs. In the lungs it picks up oxygen. The heart pumps it out to the body.

Then it comes back. That is a loop, and it never stops all day or all night.

TAKEAWAY 8W

Circulation is a loop, not a one-way trip.

END OF THE SHIFT 2W

Shift complete

THE CLOSING CARD

All six children were seen, and every one of them went home. Ava is breathing easily. Ben is walking on his ankle. The boy with the fever is asleep, with his mum in the chair next to him.

Nurse Alex Lee says you can read the clues now, and the team wants you back tomorrow.

Card text only — no options, boards or answers. 9,485 words of card prose across 58 stops. Generated from themes/hospital by tools/make-cardbook.mjs.